

Questions

Question 1	2
Question 2	3
Question 3	4
Question 4	5
Question 5	6
Question 6	7
Question 7	8
Question 8	9
Question 9	10
Question 10	11
Question 11	12
Question 12	13
Question 13	14
Question 14	15
Question 15	16
Question 16	17
Question 17	18

1. Find the exact value of the improper integral

$$\int_0^4 \frac{1}{\sqrt{x(4-x)}} \, dx \quad [5]$$

2. Show that

$$\int_0^3 \frac{1}{x^2 - 2x + 5} dx = \frac{1}{2} \arctan(3) \quad [5]$$

3. (i) Prove, from definitions involving exponentials, that

$$1 - \tanh^2 u = \operatorname{sech}^2 u \quad [3]$$

- (ii) Prove that, for $|y| < 1$,

$$\operatorname{artanh} y = \frac{1}{2} \ln \left(\frac{1+y}{1-y} \right) \quad [4]$$

- (iii) Use the substitution $x = 4 \tanh u$ to show that, for $|x| < 4$,

$$\int \frac{1}{16 - x^2} dx = \frac{1}{4} \operatorname{artanh} \frac{x}{4} + c$$

Hence deduce that

$$\int \frac{1}{16 - x^2} dx = \frac{1}{8} \ln \left(\frac{4+x}{4-x} \right) + c$$

where c is an arbitrary constant.

[6]

- (iv) By first expressing $-t^2 + 6t + 7$ in completed square form, show that

$$\int_1^5 \frac{1}{-t^2 + 6t + 7} dt = \frac{1}{4} \ln 3 \quad [5]$$

4. (i) For $x > 5$, find

$$\int \frac{1}{\sqrt{x^2 - 6x + 5}} \, dx$$

expressing your answer first in terms of arcosh and hence in logarithmic form.

[3]

- (ii) Use integration by substitution with $x - 3 = 2 \cosh u$ to show that, for $x > 5$,

$$\int \sqrt{x^2 - 6x + 5} \, dx = 2 \left(\frac{x-3}{2} \sqrt{\frac{(x-3)^2}{4} - 1} - \ln \left| \frac{x-3}{2} + \sqrt{\frac{(x-3)^2}{4} - 1} \right| \right) + c$$

where c is an arbitrary constant.

[7]

5.

$$g(x) = \frac{2x^2 - 3x + 31}{x^3 - 3x^2 + 9x + 13}$$

(a) Express $g(x)$ in the form

$$\frac{P}{x+1} + \frac{Q}{x^2 - 4x + 13}$$

where P and Q are constants to be found.

[3]

(b) Find $\int g(x) \, dx$, giving your answer in the form

$$A \ln |x+1| + B \arctan \left(\frac{x-2}{3} \right) + c$$

where A and B are constants to be found.

[4]

(c) Hence show that $\int_0^\infty g(x) \, dx$ diverges.

[2]

6. Show that

$$\int_1^4 \frac{1}{\sqrt{10+8x-2x^2}} \, dx = \frac{1}{\sqrt{2}} \left(\arcsin\left(\frac{2}{3}\right) + \arcsin\left(\frac{1}{3}\right) \right) \quad [6]$$

7. (a) Explain why

$$\int_0^{\infty} \frac{1}{x^2 - 6x + 18} dx$$

is an improper integral.

[1]

(b) Show that

$$\int_0^{\infty} \frac{1}{x^2 - 6x + 18} dx = k\pi$$

where k is a constant to be determined.

[4]

8. Evaluate

$$\int_0^4 \frac{1}{x^2 - 4x + 8} \, dx$$

giving your answer as an exact multiple of π .

[8]

9.

$$f(x) = \frac{6x + 1}{4x^2 - 8x + 13}$$

(a) Find $\int f(x) \, dx$, giving your answer in the form

$$A \ln(4x^2 - 8x + 13) + B \arctan\left(\frac{2x - 2}{3}\right) + c$$

where c is an arbitrary constant and A and B are constants to be found.

[4]

(b) Hence show that the mean value of $f(x)$ over the interval $[1, \frac{5}{2}]$ is

$$\frac{1}{2} \ln 2 + \frac{7\pi}{36}$$

[3]

(c) Hence write down the mean value of $3 - 2f(x)$ over the interval $[1, \frac{5}{2}]$.

[1]

10.

$$g(x) = \frac{4x - 1}{x^2 - 4x + 8}$$

Show that

$$\int g(x) \, dx = A \arctan\left(\frac{x-2}{2}\right) + B \ln(x^2 - 4x + 8) + c$$

where c is an arbitrary constant and A and B are constants to be found.**[5]**

11. You are given that $y = \operatorname{arcosh} x$, where $x \geq 1$.

(i) Show that

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2 - 1}} \quad [4]$$

(ii) Show, by integrating the result in part (i), that

$$y = \ln \left(x + \sqrt{x^2 - 1} \right) \quad [4]$$

(iii) Hence show that

$$\int_{15/8}^{5/2} \frac{1}{\sqrt{4x^2 - 9}} dx = \frac{1}{2} \left(\operatorname{arcosh} \frac{5}{3} - \operatorname{arcosh} \frac{5}{4} \right)$$

Express this answer in logarithmic form. [4]

12. Determine each of the following integrals.

(i)

$$\int \frac{1}{9x^2 + 12x + 20} \, dx \quad [4]$$

(ii)

$$\int \frac{1}{\sqrt{45 + 12x - 9x^2}} \, dx \quad [4]$$

13. Without using a calculator, find

(a)

$$\int_{-1/5}^1 \frac{1}{25x^2 - 20x + 13} \, dx$$

giving your answer as a multiple of π

[5]

(b)

$$\int_{-1/5}^1 \frac{1}{\sqrt{25x^2 - 20x + 13}} \, dx$$

giving your answer in the form $p \ln(q + r\sqrt{2})$, where p , q and r are rational numbers to be found.

[7]

14. (i) Given that $\tanh y = x$, where $|x| < 1$, show that

$$y = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

and hence that

$$\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) \quad [6]$$

- (ii) Use your result in part (i), or otherwise, to find

$$\int_{1/5}^{2/5} \frac{1}{4-9x^2} \, dx$$

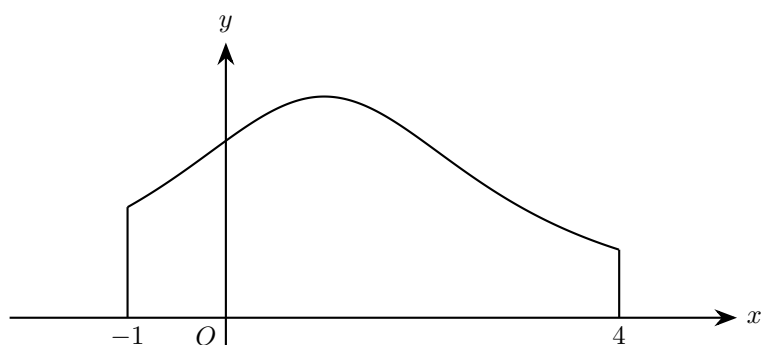
expressing your answer in an exact logarithmic form. [6]

15. Show that

$$\int \frac{x^2 + 14x + 21}{(x - 1)(x^2 + 4x + 13)} dx = A \ln |x - 1| + B \ln(x^2 + 4x + 13) + D \arctan\left(\frac{x + 2}{3}\right) + c$$

where A , B and D are constants to be found.

[7]



- 16.** The diagram shows the cross-section of a solid wooden beam.
Each unit on the axes represents 6 cm. The curved upper boundary of the beam is modelled by the equation

$$y = \frac{9}{(x-1)^2 + 4}$$

The cross-section is bounded by the curve, the x -axis and the lines $x = -1$ and $x = 4$.
Assuming that the beam is a prism of length 240 cm, find, correct to 3 significant figures, the volume of the beam.

[6]

17. Find

$$\int_1^e \frac{1}{x\sqrt{4 - (\ln x)^2}} \, dx$$

expressing your answer in terms of π .

[8]